Lab 7: Error Debugging with AI

Systematic Approaches to Finding and Fixing Bugs

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Task 1 — Syntax Error: Missing Parentheses in Print Statement

Task: Provide a Python snippet with a missing parenthesis in a print statement. Use AI to detect and fix the syntax error.

■ *Incorrect Code*

```
def greet():
    print "Hello, AI Debugging Lab!"

greet()
```

■ *Incorrect Output*

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/debug.py
  File "c:\Users\tanik\OneDrive\Desktop\c_prog\Github\debug.py", line 2
    print "Hello, AI Debugging Lab!"
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
SyntaxError: Missing parentheses in call to 'print'. Did you mean print(...)?
```

■ *Correct Code*

```
def greet():
    print("Hello, AI Debugging Lab!")

greet()
```

■ *Correct Output*

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDri
ve/Desktop/c_prog/Github/debug.py
Hello, AI Debugging Lab!
```

Task 2 — Incorrect Condition in an If Statement

Task: Supply a function where an if-condition mistakenly uses = instead of ==. Let AI identify and fix the issue.

■ *Incorrect Code*

```
def check_number(n):
    if n = 10:
        return "Ten"
    else:
        return "Not Ten"
```

■ *Incorrect Output*

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ c:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_p
rog/Github/debug.py
  File "c:\Users\tanik\OneDrive\Desktop\c_prog\Github\debug.py", line 8
    if n = 10:
       ^^^^^^
SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?
```

■ *Correct Code*

```
def check_number(n):
    if n == 10:
        return "Ten"
    return "Not Ten"


if __name__ == "__main__":
    print(check_number(10))
    print(check_number(7))
```

■ *Correct Output*

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDri
ve/Desktop/c_prog/Github/debug/task2.py
Ten
Not Ten
```

Task 3 — Runtime Error: File Not Found

Task: Provide code that attempts to open a non-existent file and crashes. Use AI to apply safe error handling.

■ *Incorrect Code*

```
def read_file(filename):
    with open(filename, 'r') as f:
        return f.read()

print(read_file("nonexistent.txt"))
```

■ *Incorrect Output*

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/debug/task3.py
Traceback (most recent call last):
  File "c:\Users\tanik\OneDrive\Desktop\c_prog\Github\debug\task3.py", line 4, in <module>
    print(read_file("nonexistent.txt"))
          ~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  File "c:\Users\tanik\OneDrive\Desktop\c_prog\Github\debug\task3.py", line 2, in read_file
    with open(filename, 'r') as f:
         ~~~~^^^^^^^^^^^^^^^^
FileNotFoundError: [Errno 2] No such file or directory: 'nonexistent.txt'
```

■ *Correct Code*

```python
# Task 3: Safe File Handling
import os

def read_file(filename):
    try:
        with open(filename, 'r') as f:
            return f.read()
    except FileNotFoundError:
        return f"Error: The file '{filename}' was not found."
    except Exception as e:
        return f"An unexpected error occurred: {str(e)}"

# Scenario 1: File Missing
print("Scenario 1 (Missing File):", read_file("nonexistent.txt"))

# Scenario 2: File Exists
temp_filename = "sample_test.txt"
with open(temp_filename, "w") as f:
    f.write("Sample content")
print("Scenario 2 (File Exists):", read_file(temp_filename))

# Clean up temporary test file
if os.path.exists(temp_filename):
    os.remove(temp_filename)

# Scenario 3: Invalid path/directory
print("Scenario 3 (Invalid Path):", read_file("/invalid_dir/nonexistent.txt"))
```

■ *Correct Output*

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/debug/task3.py
Scenario 1 (Missing File): Error: The file 'nonexistent.txt' was not found.
Scenario 2 (File Exists): Sample content
Scenario 3 (Invalid Path): Error: The file '/invalid_dir/nonexistent.txt' was not found.
```

Task 4 — Calling a Non-Existent Method

Task: Give a class where a non-existent method is called (e.g., obj.undefined_method()). Use AI to debug and fix.

■ *Incorrect Code*

```
class Car:
    def start(self):
        return "Car started"

my_car = Car()
print(my_car.drive())  # drive() is not defined
```

■ *Incorrect Output*

```
$ C:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDri
ve/Desktop/c_prog/Github/debug/task4.py
Traceback (most recent call last):
  File "c:\Users\tanik\OneDrive\Desktop\c_prog\Github\debug\task
4.py", line 5, in <module>
    print(my_car.drive())  # drive() is not defined
          ^^^^^^^^^^^^
AttributeError: 'Car' object has no attribute 'drive'
```

■ *Correct Code*

```
class Car:
    def start(self):
        return "Car started"

    def drive(self):
        return "Car is driving"


my_car = Car()
print(my_car.start())
print(my_car.drive())
```

■ *Correct Output*

```
tanik@LAPTOP-8SVR1F2C MINGW64 ~/OneDrive/Desktop/c_prog
$ C:/Users/tanik/.local/bin/python3.13.exe c:/Users/tanik/OneDrive/Desktop/c_prog/Github/debug/task4.py
Car started
Car is driving
```

Task 5 — TypeError: Mixing Strings and Integers in Addition

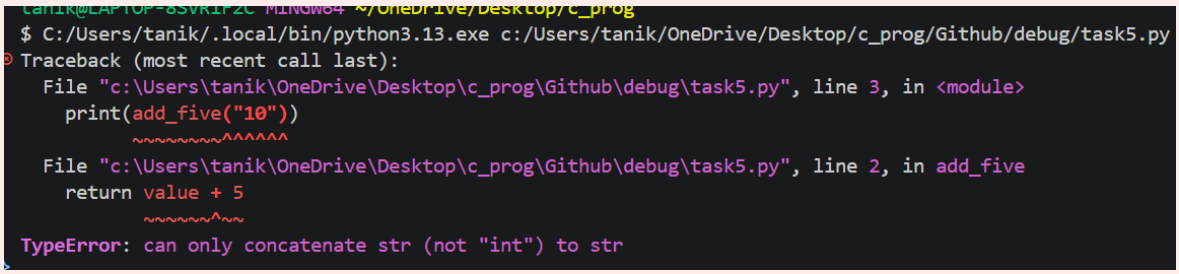
Task: Provide code that adds an integer and string ("5" + 2) causing a TypeError. Use AI to resolve the bug.

■ *Incorrect Code*

```
def add_five(value):
    return value + 5

print(add_five("10"))
```

■ *Incorrect Output*



■ *Correct Code*

```
def add_five(value):
    try:
        number = int(value)
    except (TypeError, ValueError):
        return f"Error: '{value}' is not a valid number."
    return number + 5


print(add_five("10"))
print(add_five(10))
print(add_five("abc"))
```

■ *Correct Output*

